Mining Human–AI Collaboration in Software Engineering Repositories

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ABSTRACT

Coding agents increasingly submit pull requests (PRs) alongside human developers, but it is unclear how their contributions compare. We mined the AIDev dataset, pinned to a fixed HuggingFace revision, which contains 33,596 PRs from five coding agents and 6,618 human PRs from GitHub repositories with more than 100 stars. We ask how development outcomes differ between AI-related and human PRs, and which task types are associated with AI teammates. After cleaning, we compared merge rate, time to merge, time to close, and the distribution of task types. Human PRs were merged more often than AI-related PRs (82.64% vs. 76.75% of closed PRs), while AI-related PRs were resolved much faster (median time to merge 0.05 h vs. 3.90 h). AI-related PRs were dominated by feature implementation (43% vs. 29%), documentation, and testing, whereas human PRs contained more chore and build changes. Because the dataset holds no review, commit, or file data for human PRs, our comparison is limited to PR-level outcomes and task types. The results are descriptive and do not show that AI use causes the differences.

General Terms

Artificial Intelligence, Data Mining, Data Cleaning, Empirical Software Engineering, Code Review Process, Developer Productivity, Threats to Validity, Comparative Analysis
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1. INTRODUCTION

Artificial intelligence (AI) is increasingly used in software engineering, with AI coding agents contributing alongside human developers. This study investigates human–AI collaboration by analyzing pull requests (PRs) data from GitHub repositories using software repository mining techniques. Using the AIDev dataset, we examine differences in development outcomes between AI and human contributions, as well as the types of tasks most commonly associated with AI teammates.

2. RESEARCH QUESTIONS

This study investigates the following research questions:

RQ1: How do AI-related contributions differ from human contributions in terms of development outcomes?

This question examines whether AI-related and human contributions differ in outcomes such as merge or issue closing rates, time to merge or close, number of review rounds, and revisions before acceptance.

This is important as lately more humans have been using AI to code. There are quite mixed opinions about this, with some developers using it daily, while other developers swear to avoid it.

RQ2: What types of tasks are more commonly associated with AI teammates?

This question examines which types of software engineering tasks are associated with AI-related contributions, including bug fixes, feature implementation, documentation, refactoring, testing, CI/CD or configuration, maintenance, and experimental or research-related changes.

This is important as it may be harder to distinguish certain tasks as human tasks and certain tasks as AI tasks. If AI is able to properly handle part of the code, it might spark the discussion whether AI should be used in those areas, such that the developers are able to focus on other parts of the code.

3. METHODOLOGY

3.1 Data mining

We did not crawl the GitHub API ourselves. The repository specified in the assignment, `SAILResearch/AI_Teammates_in_SE3`, is the replication package of the AIDev study [1]. It contains analysis scripts and figures, while the mined data is published as Parquet tables on HuggingFace (`hao-li/AIDev`). Our mining therefore consisted of extracting these tables, joining them on pull request (PR) and repository identifiers, and aggregating them into a single analysis-ready dataset.

We used the AIDev-pop subset, which covers repositories with more than 100 stars. It contains 33,596 PRs authored by five autonomous coding agents (OpenAI Codex: 21,799; GitHub Copilot: 4,970; Devin: 4,827; Cursor: 1,541; Claude Code: 459) in 2,807 repositories, and 6,618 human-authored PRs that serve as a baseline. The AI-related PRs were created between 24 December 2024 and 30 July 2025, and the human PRs between 1 January and 28 June 2025. The current `main` branch of the dataset no longer includes the human baseline, as the dataset was restructured in August 2026. To obtain a human comparison group and to match the replication package, we pinned the dataset to revision `68ed5f4b80d27a9e057fc57567f38bd322ac73ec` (10 May 2026), which is recorded in our manifest file so that the data collection can be reproduced exactly. For every PR we collected the PR id and number, title, description, author, agent, state, creation, closing, and merge timestamps, and repository metadata (name, primary language, stars). We added the task type of each PR, which was assigned by the dataset authors and reused without modifica-

tion. For AI-related PRs, we additionally collected comments, reviews, review comments, commits, and file-level changes. Line additions and deletions were aggregated per PR, and patch contents were not loaded. Human PRs do not contain a repository identifier, so we recovered it from the repository name in the PR URL, which succeeded for 99.26% of them (49 of 6,618 PRs keep missing repository metadata).
At the pinned revision, none of the 6,618 human PR identifiers appear in the comment, review, commit, file-level, or timeline tables, which cover only AI-related PRs. Human and AI-related PRs can therefore be compared only on the fields of the PR table itself and on task type; review participation, commit, file-level, and label metrics are available for AI-related PRs only.

3.2 Data cleaning

The (semi-)raw dataset generated by mining the data contains pull requests (PR) and columns. Cleaning was done in pandas and fully automatic and rule-based, none of the cleaning was done by hand, meaning that no manual labelling was used, the only thing that might be considered manual labelling is the filtering of "Human" out of the user types.
For removing the duplicates of the dataset it is important to look at two parts, namely the pull requests id and the html_url. A PR is identified by these two parts. This resulted in treating a row as a duplicate if and only if one of these two fields was repeated. Rows that look similar in other fashions have not been considered duplicates, as different requests can share the same text or other identifying factors.
Before dropping these however, there is a check to see how many duplicate values exists. On top of that, there is also a check whether duplicated URLs were attributed to more than one agent, to check if they were truly duplicates, or if they might simply be mislabelled. When dropping duplicates, we only kept the first instance.
Now for the missing values. The type of action was different per column. For body and labels, the missing values were filled with an empty string. These as missing values for this is a real state, and thus not an error. Number of labels, comments, reviews, review comments and reviewers were all set to zero if they were missing as we assumed that missing values mean that no such activity has been recorded in the first place. Number of files, additions, deletions, number of commits and first response at were all left as NaN as changing it to zero would lead to false values and would thus bias the averages. This would skew our analysis. Created, closed, merged and first response at were all converted to datetime, as this is needed for time-based metrics. Lastly if the task type was missing, we set the value to unknown so we can keep them in the analysis.
For the NaN columns, we compared the missing values for AI and human pull requests to check if the missingness differed per group. Furthermore, we checked the human user mean and maximum for the zero-filled count columns to see if zero was a good default. Negative directions were also set to NaN rather than dropped, so that the rest of the information in that row remained usable.
Next to that, we labeled AI vs. Human using the agent field, everything was labeled AI, unless it was labeled Human to begin with, as all other agents were large language models, or other forms of AI. We then made a column that is a binary value whether it is AI for easier analysis.
Lastly, we excluded open pull requests from the merge-rate comparison, as they are not done yet.
This resulted in a final dataset of 40158 rows.

3.3 Data classification and metrics

3.3.1 RQ1. To investigate how AI-related contributions differ from human contributions in terms of development outcomes, we analyzed the cleaned pull request dataset. These pull requests were grouped according to the contribution type field, which distinguished the AI made collaborations from the human-made ones. We compared both groups with the following metrics: merge rate, time to merge, time to close, review participation, and commits per pull request.

Merge rate was calculated as the percentage of closed pull requests that were merged, excluding open PRs whose outcomes were not yet final. Time to merge was measured as the difference between the creation and merge timestamps, considering merged PRs only. Time to close was calculated as the difference between creation and closure timestamps for all closed PRs, including both merged and unmerged PRs. Review participation was measured using the number of reviews, reviewers, and review comments per PR. Finally, commits per PR measured the number of commits associated with each pull request. However, the available review and commit records could not be matched to human PRs using the available pr id key. Therefore, these two metrics could only be analysed for AI-related PRs and could not be used for a valid comparison between the groups. Because resolution time metrics (time to merge and time to close) are non-normally distributed, we evaluated whether the differences in resolution times between AI-related and human PRs were statistically significant using the non-parametric Mann-Whitney U test at a significance level of $\alpha = 0.05$.

3.3.2 RQ2. To investigate which types of tasks are more commonly associated with AI teammates, we also analyzed the task types of the cleaned pull request dataset. Each PR was assigned a task category based on the classification provided in the original dataset, including feature implementation (feat), bug fixes (fix), documentation (docs), refactoring (refactor), testing (test), continuous integration (ci), and build-related changes (build), among others. We retained the original task categories and grouped the PRs according to their contribution type, distinguishing AI-related contributions from human contributions. All PRs were included in this analysis, regardless of their outcome.

We used the task type rate as our main metric, calculated as the percentage of PRs belonging to a specific task category within each contribution group. This metric represents the percentage of PRs associated with a specific task category out of the total PRs in the corresponding contribution group. This allowed us to identify the most common tasks performed by AI teammates and compare their distribution with human contributions, despite the difference in dataset sizes.

4. RESULTS

4.1 RQ1

The results show differences between AI-related and human pull requests in merge rates and resolution times.
Human PRs had a higher merge rate than AI-related PRs, by 5,89 percentage points. However, AI-related PRs were merged more quickly, having a median merge time of 0,05 hours, compared with 3,90 hours for human PRs. The mean time to merge was also lower for AI-related PRs, 19.43 versus 63.62 hours. A Mann-Whitney U test confirmed that the difference in time to merge between AI-related and human PRs is statistically significant ($U = 27,967,857$, $p < 0.001$).
Similarly, AI-related PRs were closed more quickly than human PRs. The median time to close was 0.12 hours for AI-related PRs,

Table 1. Comparison of development outcomes for AI-related and human pull requests.

Metric	AI-related PRs	Human PRs
Merge rate	76.75%	82.64%
Mean time to merge	19.43 h	63.62 h
Median time to merge	0.05 h	3.90 h
Mean time to close	40.45 h	93.16 h
Median time to close	0.12 h	5.37 h

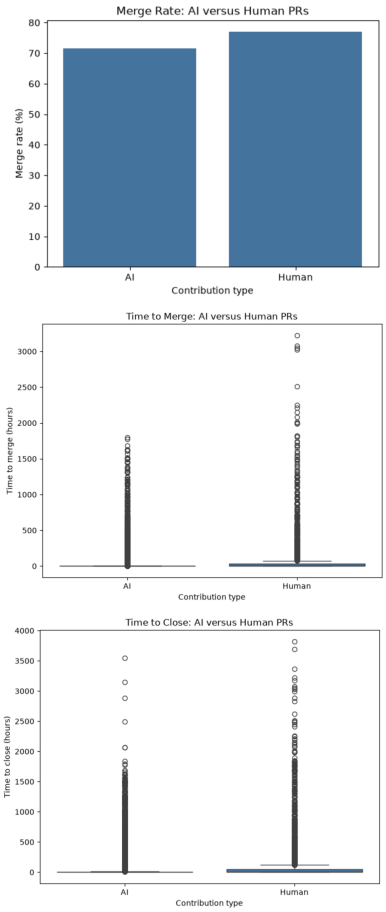


Fig. 1. Top to Bottom: merge rate, time to merge, and time to close for AI-related vs. human PRs.

compared with 5.37 hours for human PRs. The mean time to close was also lower for AI-related PRs, at 40.45 hours versus 93.16 hours for human PRs.

The mean values were much higher than the medians in both groups, which suggests that some PRs took much longer to merge or close than others.

For review participation, AI-related PRs had an average of 0.86 reviews, 0.40 reviewers and 0.80 review comments per PR. Per our data cleaning rules, unrecorded human metrics defaulted to zero due to unlinked records. Finally, for commits per PR, the available data showed that AI-related PRs had an average of 2.64 commits, a median of 1 and a range from 1 to 30 commits. In both of this cases, there was no recorded activity for humans.

4.2 RQ2

The results in the figure below show that feature implementation was the most common task type among AI-related PRs, accounting for approximately 43% of contributions. Bug fixes were the second most frequent category at 24%, followed by documentation at approximately 12%. Testing and refactoring each represented around 7% of AI-related PRs. Other categories, such as chore, build, continuous integration (CI), performance, and style, accounted for much smaller percentages, each representing less than 3% of contributions.

Overall, feature implementation and bug fixes accounted for approximately 67% of AI-related PRs. This shows that AI teammates were primarily associated with these two task categories, while other tasks, particularly configuration and maintenance-related changes, were less common in the analysed dataset.

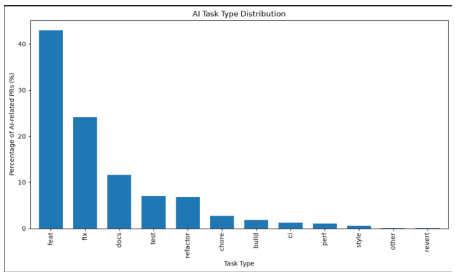


Fig. 2. AI Agents task category distribution.

5. DISCUSSION

5.1 RQ1

The results show that human PRs had a higher merge rate, while AI-related PRs were merged and closed more quickly. This means that faster resolution does not necessarily mean that a PR is more likely to be merged. The statistically significant difference in median merge times ($p < 0.001$) suggests key differences in development workflows. AI coding agents are often integrated into automated CI/CD pipelines or triggered actions that lead to near-instantaneous merges (median 0.05 h), whereas human PRs depend on human review schedules and communication (median 3.90 h).

The difference between the mean and median times also shows that, although many PRs were resolved quickly, some took much longer. These cases increased the mean values in both groups.

However, the results do not explain why these differences exist. They could be related to differences in the types of changes, their complexity or the review process, but our analysis cannot confirm this.

The review participation and commit metrics have an important limitation: the available data does not provide matching human records. Therefore, these metrics cannot be used to draw conclu-

sions about differences between AI-related and human PRs. The review values of zero for human PRs should not be interpreted as evidence that they received no reviews. Similarly, the absence of human commit records does not mean that human PRs had no commits.

Overall, the merge rate and resolution times show differences between the two groups, but they are not enough to determine which type of contribution is better in terms of code quality or overall development outcomes.

5.2 RQ2

The results show that feature implementation and bug fixes were the most common task types among AI-related PRs, accounting for approximately 67% of contributions. This suggests that AI teammates were mainly associated with implementing new features and fixing existing code in the analyzed repositories.To further understand the distribution of task types among AI-related PRs, we also compared them with human contributions to identify potential differences in the types of tasks performed by each group.

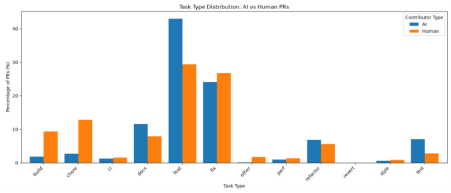


Fig. 3. AI vs Human task category distribution.

When comparing AI-related and human PRs, some differences can be observed. Feature implementation represented a higher percentage of AI-related PRs (43%) compared with human PRs (29%). AI-related PRs also had higher percentages of documentation and testing tasks. In contrast, human PRs had higher percentages of chore (maintenance) and build-related tasks, suggesting that these maintenance-related activities were more common among human contributions.

These findings suggest that AI teammates and human developers may be involved in different types of software engineering tasks. Further analysis would be needed to understand the reasons behind these differences in task distribution between AI-related and human contributions.

6. THREATS TO VALIDITY

6.1 RQ1

There are several limitations to consider when interpreting these results. First, the dataset contains more AI-related PRs than human PRs, which may affect the comparison between the two groups.

Second, the available data does not cover all metrics equally for both groups. The review and commit tables contain records for AI-related PRs, but the human PRs have no matching records using the available pr id key. This means that the review and commit metrics cannot support a valid comparison between AI-related and human PRs. In particular, the zero values for human review participation result from missing matches being filled with zeros, while the human commit records are missing and are excluded from the commit analysis.

Another limitation is that the classification of PRs as AI-related or human may not capture every case correctly. The results may also depend on the repositories included in the dataset and their development and review practices.

Finally, the analysis only considers the available PR data. We do not have enough information to explain the reasons behind each outcome, and the observed differences do not prove that the use of AI caused the differences in merge rates or resolution times.

6.2 RQ2

The analysis of AI task category distribution presents some limitations that should be considered when interpreting the results. First, the task categories were taken directly from the original dataset without manual verification. Some PRs may have been incorrectly classified, and certain contributions could belong to more than one task category.

Second, the dataset contains more AI-related PRs than human PRs. Although percentages were used to compare the two groups, the differences in task distribution may also depend on the repositories included in the dataset and the types of tasks assigned to AI agents.

Finally, the analysis only considers the frequency of each task category. It does not provide enough information to explain the reasons behind the observed differences or determine whether AI teammates perform certain tasks more effectively than humans.

With more time and data, we could manually verify the task classifications, compare contributions within the same repository, and examine whether certain task types are associated with different development outcomes.

7. CONCLUSION

This study analyzed GitHub pull requests to investigate differences between AI-related and human contributions in software engineering. T. he results showed that human PRs had higher merge rates, while AI-related PRs were merged and closed more quickly. Feature implementation and bug fixes were the most common task types among AI-related contributions.
Overall, these findings show that AI teammates and human developers differ in both the outcomes of their contributions and the types of tasks they are involved in. However, further analysis is needed to understand the reasons behind these differences. Future research could investigate the factors influencing merge rates and resolution times, as well as the factors affecting task type distributions between AI-related and human contributions.

8. AI DISCLOSURE STATEMENT

AI tools were used solely for grammar, formatting, and language clarity assistance. All data collection, cleaning, metric definition, data analysis, results generation, and final interpretations were independently conducted, reviewed, and verified by the authors."

References

[1] Hao Li, Haoxiang Zhang, and Ahmed E. Hassan. *The Rise of AI Teammates in Software Engineering (SE) 3.0: How Autonomous Coding Agents Are Reshaping Software Engineering*. 2025. arXiv: 2507.15003 [cs.SE].

APPENDIX

https://github.com/mirthe06/release_engineering_reproduction_project